



2.5 MILLION

THE NUMBER OF LIGHT YEARS **ANDROMEDA** IS FROM EARTH

Andromeda lies far beyond our own Milky Way. It has long arms spiraling out from a bulging center.

astronomer Cecilia Payne, later, Cecilia Payne-Gaposchkin (1900–79). In March, Eddington published his analysis of the **relationship between a star's mass and its luminosity** (brightness). Although a star's brightness increases with its mass, the link between the two is

not a simple one, because bigger stars have disproportionately greater luminosities. However, with Eddington's mathematical equation, it became possible to calculate a star's mass from its measured luminosity. This had important implications for understanding the life cycle of

stars, as the more massive stars have shorter lifespans.

In November, French physicist **Louis de Broglie** (1892–1987) published an idea that would revolutionize scientists' perception of atoms. He suggested that matter—like light—could have both **particle- and wavelike properties**. He devised a way to calculate the theoretical wavelength of a particle, such as an electron. He found that the wavelength values diminished to almost negligible levels for particles above the size of an atom, but for subatomic particles they were more significant. A few years later, experimental evidence would suggest that de Broglie's idea was right: electrons diffracted in wavelike ways, just like light.

At the end of 1924, more than a year after he had observed that certain stars were farther away than the other stars in the Milky Way, Edwin Hubble announced that **Andromeda was not a spiral nebula**, as was previously thought, but an entire **galaxy**,

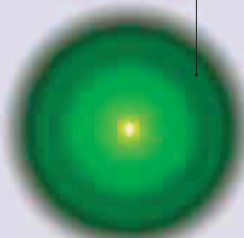
Luminous star

As a star slowly explodes, nuclear reactions at its core release energy as light. Eddington's equation made it possible to determine a star's mass based on its luminosity—in general, the brighter a star burns, the greater its mass.

WAVE-PARTICLE DUALITY

Electrons fired through a thin sheet of graphite onto a luminescent screen form a pattern of rings. These kinds of rings ordinarily arise when diffracted (scattered) waves, such as those of light, interfere with one another. This indicates that electrons, conventionally regarded as subatomic particles, must have wavelike properties too.

concentric rings on luminescent screen



equivalent to Milky Way. Andromeda stars can be more than 20 times the distance of the farthest stars of the Milky Way. The revelation settled the Great Debate of 1920—held at the Smithsonian Museum of Natural History, Washington, DC, between American astronomers

Harlow Shapley and Heber Curtis (1872–1942). Shapley had argued that the Milky Way defined the extent of the universe, but Eddington's results supported Curtis's view that a multigalactic universe was much bigger.

ARTHUR EDDINGTON (1882–1944)

Born into a Quaker family, Arthur Eddington studied at Cambridge University, UK, before pursuing a career in astronomy. He tested Albert Einstein's general theory of relativity during a solar eclipse in West Africa, and formulated theories concerning the life of stars. He was awarded a knighthood in 1930, followed by the Order of Merit in 1938.



March 5 Arthur Eddington explains the relationship between the **mass and luminosity of stars**

American physicist Wolfgang Pauli starts to develop his **Exclusion Principle**—that it is impossible for two particles to have the same quantum state

British physiologist Charles Sherrington discovers the **stretch reflex**

Indian and German physicists Satyendra Bose and Albert Einstein explain the behaviour of **fundamental force-carrying particles** that are later called bosons

October 1 British mathematician John Lennard-Jones describes **nonbonding atoms**

November 25 Louis de Broglie describes his **particle-wave theory of matter**

December 30 Edwin Hubble reports that Andromeda and the Milky Way are both **galaxies**, suggesting that the **universe is bigger** than previously thought